

**CBP-Derived Prices and Oxygen Equipment
Markets in Rural Medicare -
A Pre/Post Analysis of Modality Substitution and
Market Contraction**

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1 Medicare Program and DME

Healthcare spending has become an increasingly relevant topic for all economies in the rich world as more and more people need access to healthcare and as healthcare prices continue to increase. A large portion of this spending is made up of the medicare program, in 2024 medicare spending reached 1.1 trillion dollars, making up 21 percent of total national health expenditure. This research project focuses on a more specific branch of the Medicare program called Durable Medical Equipment (hereafter DME) and specifically how a recent policy development focused on cutting the prices of DME to reduce wastefulness in government spending.

Durable Medical Equipment (DME) refers to a specific category of reusable medical devices prescribed by the physician for home use, DME incorporates a broad range of equipment from wheelchairs, hospital beds, and respiratory equipment. DME is particularly useful for Medicare beneficiaries as it specifically facilitates usage outside of the typical primary care institutions such as inpatient care. Within the broad category of DME, supplemental oxygen occupies an important position both clinically and economically, it is the primary therapeutic intervention for patients suffering from chronic hypoxemia associated with conditions such as chronic obstructive pulmonary disease (COPD), interstitial lung disease (ILD), and heart failure, all of which are disproportionately prevalent among the elderly Medicare population. The stakes of market disruption in the oxygen equipment segment are therefore substantially high and often it may be the case that beneficiaries who lose access to this kind of care face direct threats to their respiratory function and even survival. In the next section, I will outline a major development in the pricing structure of durable medical equipment which has major consequences particularly in the market for oxygen supplies and equipment.

2 Competitive Bidding Program

2.1 Background

In 1997, the Balanced Budget Act put forth a statement that suggested using a competitive bidding structure (essentially an auction) that would reduce prices of these DME by way of this new market design. This new structure was piloted by the Centers for Medicare Services (CMS) in 2009 and fully implemented in 2011, and at this time it was given the name “Competitive Bidding Program” (CBP). The CBP initially occurred only in metropolitan areas, referred to by the CMS as Competitive Bidding Areas (CBA), but has since been expanded. In 2017 the auction pricing system was extended to areas originally without the CBP (non-CBAs) and the prices in these areas were set by transplanting prices from CBP auction outcomes in the closest nearby metro area.

This transplanting of prices introduced all sorts of market consequences which are relevant to the Medicare beneficiaries and their access to vital Oxygen equipment. Not only do rural markets face prices that are determined in a somewhat arbitrary way, but the CBP also has two main features that differentiate it from typical auctions and introduce market inefficiencies, these are the median pricing design and the caveat that winning bids are non-binding (Cramton 2011). Many authors have discussed the effectiveness of the CBP in the market for DME and come to somewhat opposing conclusions. In the following sections, I will outline theoretical evaluations of the CBP and some suggested alternatives, then I will move towards research geared towards measuring the actual implications of the CBP on prices of DME.

2.2 Theoretical Evaluation of CBP

Much of the literature surrounding the CBP auction shares the common viewpoint that this auction is inadequate for the market and, as Cramton et al. puts it: “designed to fail.” One flaw that is often pointed to in the literature is the median-pricing rule. This pillar of the CBP auction works such that after all winning bids have been selected up to a desired capacity the final market price paid to

suppliers is the median of all winning bids. Theoretically speaking, if all suppliers submitted bids at exactly their costs, this practice would immediately result in half of all of the suppliers being paid a price which is below their cost. A paper by Cramton et al. (2015) and a doctoral thesis by Dziekonska (2011) both conclude that the median-pricing rule encourages lowballing strategies because suppliers realize that a quite small bid for a product will ultimately have a miniscule effect on the final price due to this median-price rule. This expectation of lowballing leads to all sorts of other consequences such as “nonexistence of equilibrium” as well as allocative inefficiencies due to high-cost firms replacing lower cost firms as suppliers (Winegarden, 2018).

Another commonly touted issue with the CBP auction design is the fact that suppliers that win the auction and have their bid accepted are not obligated to sign a contract for that product. The lack of a binding commitment allows firms to again submit lower bids than they would otherwise in order to get a contract option with the knowledge that they are not obligated to supply the product at the final market price (Cramton, 2011). This market flaw along with a median-pricing rule means that if the final price for a certain product is too high for a suppliers liking they can immediately revoke their bid, thus the auction can have problems of quantitative inefficiency in which the market demand is not entirely met (Cramton, 2011 and Cramton et al., 2015, and Merlob et al., 2010). It is also true that the strategy of lowballing (sometimes called “suicide bidding”) again results in more allocative inefficiencies because low cost suppliers who submit truthful bids will be expected to supply at a much lower price than what is reasonable for the market due to many other suppliers suicide bidding in order to win contract options (Ji, 2023).

3 Empirical Findings

3.1 Utilization Trends and Supply Shortages

Based on empirical data of Medicare claims, the implementation of the CBP resulted in a measured decline in DME usage, although the extent to the decline is debated by researchers. That being said,

much of the literature seems to insist that while the program did indeed reduce prices dramatically as expected, it also created supply shortages and not merely surplus demand reductions (Ji, 2023). A study conducted by Yunan Ji in 2023 estimates that the price of DME fell by 45% on average as well as an 11% reduction in utilization across the board. Ji concludes that the CBP corresponded to a movement down on the supply curve in which the prices were set below market-clearing levels. Similarly, Ding et al. (2025) finds that the spending on a specific DME: CPAP machines, fell by 47.2%. Survey data from Clark et al. (2024) and Winegarden (2018) find that suppliers were especially affected and often had to delay deliveries for specialized equipment such as liquid oxygen, which only accounted for 2.9% of high-flow oxygen claims in Pennsylvania.

3.2 Clinical Outcomes, Mortality Rates, Unplanned Hospitalizations

Another way to study the potential market failures that could arise as a result of the flaws of the CBP auctions is by looking at patient outcomes and data that measures how useful these DME products are in helping those for whom they are intended. A study done in 2022 by Tarazi et al. uses an extended time-series data analysis method that argues the implementation of the CBP reduced prices and did not result in an increase in adverse health effects such as visits to the ER or hospitalizations, however, these findings are specific to metropolitan CBAs. This finding was reiterated by Duan et al. (2024) who found that the CBP did not result in differential changes in all-cause mortality nor Chronic Obstructed Pulmonary Disease (COPD)-related hospitalizations.

On the other hand, Cramton et al. (2012) and Winegarden (2018) find that in pilot program data the decreases in utilization result in an increase in the risk of death, duration of hospitalization, and frequency of ER visits. Some researchers have made the argument that the reason for the discrepancy in the literature is due to the fact that utilization declines were heavily concentrated in the less medically appropriate groups such as those without formal diagnoses, this would mean that studies focusing solely on claims made by those with ailments such as COPD would miss out on where the utilization decrease was actually occurring (Ding et al., 2021).

4 Modality Substitution and Geographical Access

4.1 Evidence of Modality Substitution

A vast majority of the literature discussed in the previous sections takes quite a zoomed out approach in evaluating the effect of the CBP and its auction pricing on DME. The study by Clark et al. (2024) looking at data from Pennsylvania looks specifically at oxygen as a subsection of DME, in this study the researchers find that producers began to withdraw supply because of certain reimbursement structures that failed to account for the high operational costs of liquid oxygen (LOX) systems. Between the years 2013 and 2019 the percentage of beneficiaries using LOX fell by 89% for stationary units and 86% for portable units; in addition, among patients that require high-flow delivery of oxygen (LOX is clinically approved for this group) only 2.9% of portable claims were for LOX systems (Clark et al.). As LOX has been taken out of the market, suppliers have been replacing these systems with more profitable but less effective alternatives. These suppliers have begun to provide more stationary concentrators and heavy compressed oxygen tanks and according to Clark et al. (2024) the former of these account for almost 94% of claims. This study takes a comprehensive overview of data from Pennsylvania but there is reason to believe that this trend occurred all over the US and in particular in rural areas. It is because of this that investigating modality substitutions seems an interesting next step towards having a holistic understanding of the effects of the CBP on DME.

4.2 Modality Substitution in Non-CBAs

There are a few reasons why modality substitution in a theoretical sense should be particularly acute in non-CBAs which motivate this research project. For one, rural suppliers face structural cost disadvantages relative to their urban counterparts due to long delivery routes and the sparsity of beneficiaries in these areas. Under these conditions, a profit-maximizing supplier facing downward price pressure has a strong incentive to deprioritize liquid oxygen systems in favor of concentrators, exactly the kind of modality substitution found in Clark et al. Critically, however,

this substitution is not clinically neutral. Portable and stationary concentrators have maximum output limitations that render them inadequate for patients requiring continuous high-flow oxygen, and may be unsuitable for patients with severe hypoxemia, limited mobility, or unreliable electricity access. These characteristics are disproportionately prevalent in the rural Medicare population. Supplier-driven substitution toward lower-cost modalities may therefore produce a systematic mismatch between delivered equipment and clinical need, a form of implicit undertreatment that would not be captured in aggregate utilization measures but could nonetheless contribute to preventable deterioration in respiratory outcomes among these vulnerable rural beneficiaries.

5 Finalized Research Questions

Given the following background and literature overview, there are a few different interesting research questions that have yet to be satisfyingly answered regarding the market consequences as a result of the CBP. In particular, studies like Tarazi et al. looking at the market and clinical outcomes of the CBP on metropolitan areas find little to no effects, but this study mechanically does not include rural areas where there is plenty of potential for market consequences. Therefore, the following question comes about naturally: Did the introduction of CBP-derived reimbursement rates in non-CBA markets alter the distribution of oxygen delivery modalities, supplier participation, and total claim volume among Medicare beneficiaries in rural states following the policy's implementation? In a similar light, Clark et al. studying the effects of the CBP on oxygen supplies finds evidence of modality substitution but only in a limited sample, in light of this, an important issue yet to be resolved would be to ask if these changes consistent with a shift toward lower-cost, portable oxygen delivery modalities and away from higher-capacity stationary systems, a pattern indicative of supplier-driven modality substitution under price pressure? These are the two foundational research questions that anchor this research paper.

6 Data

6.1 Primary Dataset

The primary data source is the Medicare Provider Utilization and Payment Data: Equipment, Devices, and Supplies — by Geography and Service which is a Public Use File (PUF), produced annually by the Centers for Medicare and Medicaid Services (CMS). This dataset contains HCPCS-level claim counts, beneficiary counts, and supplier counts, for the Medicare Program, organized by state and procedure code. Five annual files are employed corresponding to calendar years 2015–2019 yielding a state-level panel dataset. The unit of observation is one HCPCS code in one state in one year. The three dependent variables of interest here: beneficiary counts, suppliers counts, and claim counts, are all meaningful to understanding the impact of the price change in different ways. The term “HCPCS” means Healthcare Common Procedural Coding System and in the context of claim counts it gives an interesting measure of the demand in a given state for each individual piece of equipment, these can be sorted into high- and low-cost modalities and give interesting purview into how the price changes affect the demand-side of the market. Beneficiary counts in this dataset are a precise measure of the number of Medicare enrollees that used any kind of oxygen equipment, this too can be aggregated at the HCPCS level to investigate the comparison between high- and low-cost modalities. The last dependent variable of interest for this study is the supplier counts data, this is a fairly straightforward measure of the number of suppliers for a certain type of equipment in a state, like the other two this metric can also be aggregated at the claim level for all of the different oxygen modalities.

The five-year study window is chosen to correspond precisely to the regulatory events of interest. Calendar year 2015 represents the pre-treatment baseline, during which all non-CBA areas were reimbursed under the unadjusted traditional DMEPOS fee schedule. Calendar year 2016 captures the partial treatment period, during which a 50/50 blended rate was phased in for non-CBA areas. Finally, calendar years 2017–2019 represents the full treatment years, during which the completely adjusted CBP-derived fee schedule took effect on January 1, 2017.

6.2 Sample Restriction

Because the dataset is aggregated at the state level, states containing large metropolitan competitive bidding areas introduce contamination into state-level outcome measures; in order to mitigate this, the analysis is restricted to states in which the majority of DMEPOS ZIP codes carry the rural non-CBA designation.

Rural status at the ZIP code level is determined using the CMS DMEPOS Rural ZIP Code designation file (Q4 2015 quarterly update), which classifies each U.S. postal code as rural or non-rural according to the regulatory definition at 42 CFR §414.202: a geographic area represented by a ZIP code in which at least 50 percent of the total geographic area is estimated to fall outside any Metropolitan Statistical Area, as well as any low-population-density ZIP code within an MSA that has been excluded from a competitive bidding area.

States are included in the analytical sample if at least 60 percent of their total ZIP codes carry the rural DMEPOS designation, this is a slightly arbitrary cutoff but nonetheless restricts the data into a more straightforward subset. The 60% threshold yields a sample of 15 states, presented in Table 1 below. These states span four major geographic regions of the United States: the Mountain West, Northern Plains, the South, and New England, providing cross-regional diversity that reduces the likelihood that findings are attributable to any single regional factor or small area variation.

Table 1: Analytical Sample: States with $\geq 60\%$ Rural DMEPOS ZIP Code Share (Q4 2015)

Rank	State	Rural ZIPs	Total ZIPs	Rural ZIP Share
1	Vermont	258	270	95.6%
2	West Virginia	462	569	81.2%
3	Montana	334	415	80.5%
4	Wyoming	163	204	79.9%
5	North Dakota	328	427	76.8%
6	Mississippi	388	513	75.6%
7	Kentucky	635	848	74.9%
8	South Dakota	305	407	74.9%
9	Arkansas	454	617	73.6%
10	Maine	314	449	69.9%
11	Iowa	697	1,000	69.7%
12	Nebraska	428	637	67.2%
13	Oklahoma	457	685	66.7%
14	Kansas	476	747	63.7%
15	New Mexico	285	454	62.8%

Source: CMS DMEPOS Rural ZIP Code File, Q4 2015. Rural ZIP share calculated as rural DMEPOS ZIP codes divided by total state ZIP codes.

7 Hypothesis on Dependent Variables

7.1 Supplier Counts

The implementation of CBP-derived prices will naturally decrease prices in the market. Since these price changes are transplanted from urban CBAs it is quite possible that suppliers in the rural markets will no longer be able to keep their production efficient. As a result many suppliers may exit

the market entirely rather than continuing to earn less. Another possible scenario is that suppliers choose to discontinue the production of their high-cost equipment that require regular deliveries across large areas, in this case they may instead choose to only supply lower cost equipment and so even if supplier counts stay high there may be a market shift towards lower-cost equipment. The expectation remains the same and that is of course that after the CBP-derived prices were implemented, what should happen is that supplier counts drop precipitously across the states in the sample.

7.2 Beneficiary Counts

Following from the logic of the supplier counts, a similar argument can be made about the number of unique beneficiaries receiving oxygen equipment following the implementation of CBP-derived prices. In particular, rural markets do not have a large quantity of available suppliers and as a windfall of suppliers exiting the market beneficiary counts should similarly decrease. One important note to make about this outcome is that declining beneficiary data is a result of 3 possible consequences, either a particular beneficiary stopped receiving equipment, they have switched to a different HCPCS code, or they have left the Medicare program altogether. This makes it particularly useful as a dependent variable for studying the secondary research question of modality substitution, in particular if this hypothesis about suppliers phasing out high-cost equipment in exchange for lower cost equipment is true then we would expect to see a decrease in the number of beneficiaries for high-cost equipment and vice versa for the lower cost equipment.

7.3 Claim Counts

The third and final dependent variable in this study is assessing the number of claims submitted across all oxygen equipment in the rural state sample. The claims data is the most straightforward and interpretable of the three dependent variables being assessed here, and given the expectations outlined for the number of beneficiaries and suppliers I expect that the same applies also to the number of claims for these rural states. As a downstream consequence of supplier counts and the

number of beneficiaries decreasing it is reasonable to assume that the number of claims will also decrease.

Table 2: Summary Statistics by Year: Claims, Beneficiaries, and Suppliers

Year	Claims		Beneficiaries		Suppliers	
	Mean	SD	Mean	SD	Mean	SD
2015	6,821.99	15,188.06	1,200.74	2,254.10	85.74	89.57
	$N = 151$		$N = 150$		$N = 151$	
2016	6,720.32	15,701.56	1,177.03	2,325.35	81.06	90.44
	$N = 151$		$N = 148$		$N = 151$	
2017	6,590.00	15,403.74	1,149.66	2,316.88	77.64	89.79
	$N = 151$		$N = 149$		$N = 151$	
2018	6,355.85	14,978.01	1,156.20	2,302.05	71.91	83.75
	$N = 151$		$N = 143$		$N = 151$	
2019	6,173.64	14,131.23	1,129.20	2,160.57	67.94	80.81
	$N = 151$		$N = 138$		$N = 151$	

Note: Unit of observation is one HCPCS code in one state in one year. Claims and beneficiary counts are suppressed by CMS for cells with fewer than 11 beneficiaries, accounting for variation in N across outcomes and years. SD = standard deviation.

8 Analysis and Results

What is immediately noticeable about the summary statistics for the dependent variables is that all three proxies for how the market responded to this price cut align with the general story of a market contraction outlined in the hypothesis section. The mean claims across this time span decreased steadily, in 2015 this number was 6,822 which decreased to 6,174 in 2019, a roughly 10% reduction over the 4 year time span. Supplier counts follow a similar trend going from about 86 in 2015 to 68

in 2019, suggesting that a meaningful number of suppliers did indeed exit the market following the CBP-derived price implementation. Beneficiary data corroborates this idea of a market contraction, albeit declining only modestly, but nonetheless going from 1,201 in 2015 to 1,129 in 2019. These descriptive pre/post trends are consistent with the idea of a market contraction following a major change in the prices for oxygen equipment.

8.1 Baseline Regression

In order to really investigate this hypothesis and come to a more causal conclusion I chose to run a pooled pre/post OLS regression with clustered standard errors, or also called a naïve interrupted time-series that uses a single binary treatment variable of the CBP-derived price implementation. The theoretical equation for this regression is as follows:

$$Y_{s,t} = \alpha + \beta_1 \text{Post}_t + \varepsilon_{s,t} \quad (1)$$

where $Y_{s,t}$ denotes the outcomes of interest in states s for years t . The three outcomes are estimated separately: total Medicare claims (*clms*), total unique beneficiaries (*benes*), and total active suppliers (*suplrs*), each measured by the standard unit of measurement for this study of one HCPCS procedure code in one state in one calendar year. The variable Post_t is a binary indicator equal to one for calendar years 2017, 2018, and 2019 which are the three years of fully adjusted CBP-derived prices in non-CBAs and equal to zero for calendar year 2015, the pre-treatment baseline. Calendar year 2016, during which a transitional 50/50 blended rate was in effect, is excluded from this baseline specification to produce a clean pre/post contrast but it is reintroduced in the augmented specification described below using a partial binary treatment variable. The coefficient $\hat{\beta}_1$ therefore captures the average change in each outcome associated with full CBP-derived price implementation relative to the 2015 baseline.

Table 3: Baseline Regression Results: Market Contraction Following CBP-Derived Price Implementation (Equation 1)

	(1)	(2)	(3)
	Claims	Beneficiaries	Suppliers
Post	−448.821*** (114.841)	−55.473** (22.468)	−13.236*** (2.444)
Constant	6,821.987*** (1,038.406)	1,200.740*** (180.175)	85.735*** (10.662)
Observations	604	580	604
Clusters	15	15	15
<i>F</i> -statistic	15.27	6.10	29.32
<i>R</i> ²	0.0002	0.0001	0.0044
Root MSE	14,909	2,256.7	85.993

Notes: Dependent variables are total Medicare claims (1), total unique beneficiaries (2), and total active suppliers (3), each measured per state-HCPCS cell per year. *Post* is a binary indicator equal to one for calendar years 2017–2019 and zero for the 2015 pre-treatment baseline. Calendar year 2016 is excluded from this specification. Standard errors in parentheses are clustered at the state level (15 clusters). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Standard errors are clustered at the state level to account for the non-independence of observations within states across HCPCS codes and years. All three outcome equations are estimated separately rather than jointly, as the outcomes are conceptually distinct stages of the supply chain: supplier entry, beneficiary access, and service volume, pooling them into a single system is not warranted by the research design. Identification relies heavily on the assumption that, absent the 2017 price change, the outcomes would have remained mostly stable relative to the 2015 baseline. Because claims, beneficiaries, and suppliers all represent distinct stages of the supply chain and market, three separate regressions are run for each distinct dependent variable rather than a single stacked

model. This approach also facilitates the kind of code-specific subgroup analysis that is crucial for answering the secondary research question without imposing analytical restrictions.

The results of this initial regression specification across the full analytical sample yield statistically significant post-period declines for all three dependent variables, consistent with the broad market contraction hypothesis. Total Medicare claims in these rural states fell by an estimated 448.8 claims per state-HCPCS cell in the post period relative to the 2015 baseline ($SE = 114.84$, $p = 0.002$), representing a reduction of approximately 7%. Total unique beneficiaries declined by 55.5 per state-HCPCS cell relative to a baseline mean of 1,201 ($SE = 22.47$, $p = 0.027$), which is a reduction of almost 5% and suggestive of constrained beneficiary access. The largest and most precise estimate was observed in supplier count data, which fell by 13.2 suppliers per state-HCPCS cell relative to a baseline mean of 85.7 ($SE = 2.44$, $p < 0.001$), corresponding to a decline of approximately 15.4 percent and indicating meaningful supplier exit from non-CBA markets following the change in prices. Taken altogether, these results wholly align with the narrative that downward pressure in prices reduced profitability for firms, leading them to exit, and resulting in a contraction in the market that had downstream consequences for both beneficiary access and the number of overall claims submitted. That being said, this pooled estimate model does not yet speak to the compositional dimension of the research question which specifically asks whether the reduction in aggregate utilization was accompanied by a shift in the supply of certain modalities toward lower-cost equipment categories.

Table 4: Robustness Check: Augmented Specification with Partial Treatment Year (Equation 2)

	(1)	(2)	(3)
	Claims	Beneficiaries	Suppliers
Partial	−101.662 (72.389)	−23.713 [†] (12.338)	−4.675*** (0.862)
Post	−448.821*** (114.898)	−55.473** (22.479)	−13.236*** (2.446)
Constant	6,821.987*** (1,038.923)	1,200.740*** (180.267)	85.735*** (10.667)
Observations	755	728	755
Clusters	15	15	15
<i>F</i> -statistic	7.63	4.04	17.30
R^2	0.0002	0.0001	0.0041
Root MSE	15,070	2,270.8	86.898

Notes: Dependent variables are total Medicare claims (1), total unique beneficiaries (2), and total active suppliers (3), each measured per state-HCPCS cell per year. *Partial* is a binary indicator equal to one for calendar year 2016, during which a transitional 50/50 blended reimbursement rate was in effect. *Post* is a binary indicator equal to one for calendar years 2017–2019. The 2015 pre-treatment baseline is the omitted category. All years 2015–2019 are included in this specification (N increases relative to Table 2). Standard errors in parentheses are clustered at the state level (15 clusters). [†] $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

8.2 Robustness Check — Secondary Regression Specification

As a robustness check for these results a second pooled regression can be run using the same clustered standard errors but this time including a coefficient for the partial price change that took place in 2016 where the price was set exactly by half of the new CBP-derived prices and half of the old fee-schedule prices. The augmented specification takes the following form:

$$Y_{s,t} = \alpha + \beta_1 \text{Partial}_t + \beta_2 \text{Post}_t + \varepsilon_{s,t} \quad (2)$$

The results of this robustness check provide some corroborating evidence that are consistent with the baseline results. For total claims, the partial treatment coefficient is negative but statistically insignificant ($\hat{\beta}_1 = -101.66$, $\text{SE} = 72.39$, $p = 0.182$), while the full post-period coefficient remains large and precisely estimated ($\hat{\beta}_2 = -448.82$, $\text{SE} = 114.90$, $p = 0.002$), indicating that claim volume did not respond meaningfully to the transitional blended rate, but then contracted sharply upon full CBP-derived price implementation in 2017. The beneficiary results follow a similar pattern, with a marginally insignificant partial coefficient ($\hat{\beta}_1 = -23.71$, $\text{SE} = 12.34$, $p = 0.075$) and a statistically significant full post-period decline ($\hat{\beta}_2 = -55.47$, $\text{SE} = 22.48$, $p = 0.027$), suggesting that beneficiary access began to erode during the transition year but that the principle contraction occurred only after the full implementation. The supplier data shows meaningful deviations from this pattern, the partial coefficient is statistically significant and negative ($\hat{\beta}_1 = -4.68$, $\text{SE} = 0.86$, $p < 0.001$), meaning that suppliers began to exit the market as early as 2016 when the blended-rate phase-in took place, a finding that could very well be explained by forward-looking supplier behavior under anticipated price pressure. The coefficient for the post variable in this new specification also stayed at its original value in this regression specification ($\hat{\beta}_2 = -13.2$, $\text{SE} = 2.44$, $p < 0.001$). In all three outcome equations, the partial coefficient is smaller in absolute value than the post coefficient, satisfying the a priori prediction that $|\hat{\beta}_2| > |\hat{\beta}_1|$.

Together these results and the results from the primary regression specification support the market contraction interpretation outlined in the hypothesis section. The robustness check in particular corroborates these findings and demonstrates that the results of the original specification arose discontinuously from the fee schedule change itself rather than from any pre-existing secular trend. This is motivated by the fact that the partial coefficients follow the same trend but have smaller magnitudes, precisely what a supply-side price pressure story predicts, and it is the finding that motivates the further analysis on modalities that follows.

Table 5: Regression Results: Low-Cost Modality

	(1)	(2)	(3)
	Claims	Beneficiaries	Suppliers
Partial	666.333*** (188.841)	77.667*** (20.879)	4.600** (1.826)
Post	3,837.333*** (586.997)	537.667*** (83.879)	17.378*** (3.689)
Constant	7,299.067*** (919.338)	1,066.267*** (132.084)	95.933*** (12.377)
Observations	75	75	75
Clusters	15	15	15
F -statistic	21.37	20.74	11.11
R^2	0.1119	0.1132	0.0246
Root MSE	4,972.8	701.78	48.334

Notes: Dependent variables are total Medicare claims (1), total unique beneficiaries (2), and total active suppliers (3). *Partial* is a binary indicator equal to one for the partial treatment year. *Post* is a binary indicator equal to one for the post-treatment period. Standard errors in parentheses are clustered at the referral provider geographic code level (15 clusters). ** $p < 0.05$, *** $p < 0.01$.

Table 6: Regression Results: High-Cost modality

	(1)	(2)	(3)
	Claims	Beneficiaries	Suppliers
Partial	−261.071*** (32.966)	−37.929*** (4.920)	−6.000*** (0.752)
Post	−623.167*** (78.130)	−73.437*** (13.895)	−15.476*** (2.123)
Constant	1,038.143*** (197.698)	144.643*** (29.807)	30.000*** (3.963)
Observations	70	62	70
Clusters	14	14	14
<i>F</i> -statistic	37.95	30.64	34.40
R^2	0.1541	0.0991	0.2549
Root MSE	610.81	92.573	11.185

Notes: Dependent variables are total Medicare claims (1), total unique beneficiaries (2), and total active suppliers (3). *Partial* is a binary indicator equal to one for the partial treatment year. *Post* is a binary indicator equal to one for the post-treatment period. Standard errors in parentheses are clustered at the referral provider geographic code level (14 clusters). *** $p < 0.01$.

8.3 Investigation of Modality Substitution

Moving on toward the secondary research question of modality substitution, in order to test this hypothesis equation 1 of the original regression specification is re-estimated separately for HCPCS code E0439 (liquid oxygen portable system) and E1392 (stationary concentrator with portable add-on). These two codes that have been selected are the a priori primary empirical proxies for high- and low-cost oxygen delivery modalities respectively; divergent coefficient signs across these

codes would indubitably give direct evidence of supplier-driven modality substitution under price pressure.

The results are strikingly divergent across the two modalities and fully consistent with supplier-driven modality substitution under price pressure. For E0439, the high-cost modality, claims declined by 261.1 per state-code cell during the partial treatment year ($SE = 32.97$, $p < 0.001$), a reduction of approximately 25.1 percent relative to the 2015 baseline mean of 1,038. This value dropped even further to 623.2 claims in the full post-period ($SE = 78.13$, $p < 0.001$), this ultimately demonstrates a cumulative decline of almost 60%. Beneficiary and supplier data followed nearly identical trends, with beneficiaries falling 37.9 ($SE = 4.92$, $p < 0.001$) and 73.4 ($SE = 13.89$, $p < 0.001$) in the partial and post periods respectively, corresponding to declines of 26.2 and 50.8 percent. The number of suppliers contracted by 6.0 ($SE = 0.75$, $p < 0.001$) and 15.5 ($SE = 2.12$, $p < 0.001$), or 20.0 and 51.6 percent relative to the 2015 baseline mean of 30.

For E1392, the low-cost modality, the coefficients reverse sign entirely as claims increase by 666.3 in the partial period ($SE = 188.84$, $p = 0.003$) or a 9.1 percent relative to a baseline of 7,299, before rising a further 3,837.3 in the full post-period ($SE = 587.00$, $p < 0.001$), a cumulative increase of roughly 52.6 percent. Beneficiaries grew by 77.7 ($SE = 20.88$, $p = 0.002$) and 537.7 ($SE = 83.88$, $p < 0.001$) across the two periods which are gains of 7.3 and 50.4 percent against a baseline of 1,066. Lastly, the number of suppliers even expanded by 4.6 ($SE = 1.83$, $p = 0.025$) and 17.4 ($SE = 3.69$, $p < 0.001$), or 4.8 and 18.1 percent relative to a baseline of 95.9.

These findings demonstrate that the opposing trajectories across all three outcome measures, a contraction in the high-cost modalities and expansion in the low-cost modality both scaling nicely to the price changes, give direct empirical evidence for supplier-driven modality substitution. This evidence can go further to suggest that a meaningful share of the aggregate market contraction documented in the baseline results reflect not simply an exit from the oxygen equipment market altogether, but rather their strategic reorientation toward lower-cost, lower-margin equipment categories as price pressure intensified.

9 Conclusions, Limitations, and Next Steps

9.1 Discussion

There are several notable limitations and constraints of this research that warrant acknowledgement, firstly, state-level aggregation precludes direct identification of rural versus urban beneficiaries within each state; the sample restriction to states with a high portion of people in non-CBAs mitigates but does not fully resolve this concern, as even the most rural states in the sample contain some metropolitan ZIP codes with distinct market dynamics. The data source used for this research project comes from the CMS and there exist certain cell suppression rules for privacy including one suppresses observations with fewer than 11 beneficiaries, this may systematically undercount claims in the most sparsely populated rural areas, biasing estimates of access restriction toward zero and likely causing the analysis to understate, rather than overstate any adverse effects of the policy change.

On another note, the time period provides limited statistical power for models incorporating multiple control variables, and the absence of pre-2015 data prevents formal pre-treatment testing which would demonstrate clearly the kind of discontinuity that is expected. This also means that for my secondary hypothesis about the modality substitution there is no way of truly verifying that the two modalities were not already following these divergent trends, thus the true effect may not be causally linked to the price changes which would undermine my interpretation made in the conclusions section.

A main limitation that lies more in the realm of the chosen methodological strategy of this research is in the fact that there is no clear causal identification strategy due to all 15 states being treated simultaneously. Opting for a pre/post comparison like the strategy in this paper means that there is no true control group and so a true difference-in-differences design is impossible in reality. A significant limitation is therefore that a pre-existing or entirely confounding trend may be driving the changes that are contemporaneous, but not caused by, the CBP-derived price changes. In

addition to the treatment problem there is also a major limitation just in the specification of the regression equation because no covariates like population, health status, or income are directly controlled for across states and so changes in these factors across states in the sample could be partially driving observed patterns in the results. A future study could advance on my research by and investigate this effect more causally by using controls for fixed effects at the state-level and potentially a dataset with a more comprehensive timeline.

Lastly, the two hypotheses mainly show strong evidence about market outcomes and dynamics as a result of the price changes and do not empirically engage with health outcomes in any meaningful way. Without some data taken directly from MedPAR or a similar source, the clinical significance of the compositional shift remains an open and important question to be resolved. This leaves some interesting opportunities for future research, either with a better dataset including more observations and clinical outcomes data or a better regression specification including fixed effects at the state-level.

9.2 Conclusion

The findings of this study provide clear and substantive empirical support for both research questions posed at the outset. For the first of these two research questions: whether CBP-derived fee schedule implementation altered the distribution of oxygen delivery modalities, supplier participation, and total claim volume among Medicare beneficiaries in rural states, the evidence is unambiguous, all three measured outcomes contracted significantly following full CBP-derived price implementation. Claim volume falling by approximately 6.6 percent, beneficiary access declining by 4.6 percent, and supplier participation contracting by 15.4 percent relative to the 2015 baseline, the augmented specification confirms that these contraction scaled with the depth of the price cut, emerging modestly during the 2016 blended-rate transition and intensifying sharply upon full implementation. The second and even more consequential question, whether these changes could be thought of as a supplier-driven modality substitution toward lower-cost equipment, the disaggregated results show an equally definitive answer: the amount of claims, beneficiaries, and suppliers

associated with the high-cost stationary liquid oxygen system (E0439) declined across the board by over 50% for each metric over the study window. The corresponding figures for low-cost equipment rose across the board by as much as 52% over the same period, this striking divergence is difficult to attribute to anything other than deliberate supplier reorientation under price pressure.

Taken together this study illuminates how the CBP-derived fee schedule achieved a portion of its cost savings not through efficiency gains or the elimination of clinically unnecessary utilization, but through a compositional shift in the equipment mix toward modalities that are simply cheaper to supply. This may carry a great deal of truly unexamined consequences for the subset of rural Medicare beneficiaries for whom a stationary high-flow oxygen system is medically appropriate but potentially unavailable. This finding is particularly significant in light of Tarazi et al. (2022), whose MSA-level design mechanically and intentionally excludes the rural non-CBA populations examined here, and whose null result on adverse health outcomes may therefore reflect the geographic scope of the analysis as much as it does the true impact of the policy.

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